

$$u_{tt} = c^2 u_{xx}, \qquad 0 < x < L, \quad t > 0,$$

$$u(0, t) = u(L, t) = 0, \qquad c > 0,$$

$$u(x, 0) = A \sin(\pi x / L), \qquad u_t(x, 0) = 0,$$

$$\xi = x / L, \qquad \theta = ct / L, \qquad v = u / A,$$

$$v_{\theta\theta} = v_{\xi\xi}.$$